

Redrawing the Lines: Did Political Incumbents Influence Electoral Redistricting in India?*

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Abstract

In 2008, the boundaries of state electoral constituencies in India were redrawn for the first time since 1976. Using publicly available information, including India's census, as well as GIS and electoral data from 23 states, we provide evidence that India's delimitation process was largely politically neutral. We construct three measures of redistricting (delimitation) in a given constituency that may be unfavorable to incumbents: the extent of demographic change, the share of new voters and whether there was a change in the reservation status for minority group representation. We find that, on average, constituencies represented by advisory members of the delimitation commission, state ministers, and members of the state ruling party did not have any preferential redistricting. In a few states, advisory committee members and state ministers are significantly less likely to see reservation status change in their constituency.

Keywords: electoral redistricting, incumbency advantage, political reservations, Delimitation Commission, political influence, recontest.

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1. Introduction

Most democratic countries redraw their electoral boundaries every few years, with the goal of equalizing constituency population sizes based on the “one person one vote” principle. This process can be seen as undemocratic if influenced by incumbent politicians or ruling parties to their advantage (a process sometimes called “gerrymandering”). The largely United States-focused literature finds that partisan redistricting can shape incumbent name recognition, likelihood of post-redistricting voting, policy moderation and politician race entry (Cain 1985; Gelman & King 1994; Baqir 2002; Engstrom 2006; Besley and Preston 2007; Hayes & McKee 2009; Williamson 2019; Fraga, Moskowitz & Schneer 2022). The effect of redistricting on incumbency advantage remains contested, with some studies finding increased incumbency advantage (Gilligan & Matsusaka 1999; Shotts 2001; Forgette & Winkle 2006; Gul & Pesendorfer 2010) and others finding none (Friedman & Holden 2009).

Electoral redistricting in India (“delimitation” or “re-delimitation,” locally) has generated only limited scholarship (Singh 2000; McMillan 2000; Verma 2002; Jensenius 2013; Patel & Sekher 2024; Mishra 2025), despite India redrawing national and state electoral constituency boundaries in 2008 after a three-decade gap. While a non-political Delimitation Commission conducted the redistricting, incumbent politicians may have still been able to influence the process: several were part of the Commission’s Advisory Committee and the literature suggests other possible paths to influence (Rossiter, Johnson & Pattie 1997; Thomas 2011).

We examine whether incumbent politicians were able to exert influence on India’s redistricting process, using detailed, constituency-level demographic and electoral data to construct three constituency-level measures of the extent of redistricting (delimitation). The first two are the extent of change in a constituency’s demographic characteristics and the fraction of

new voters in the redrawn constituency, based on the assumption that such changes are likely to impose greater re-election campaign costs and are therefore adverse outcomes from an incumbent politician's perspective. A third measure is change in a constituency's constitutionally-provided-for reservation status (from unreserved to reserved for particular social groups¹ or vice versa). When reservation status changes, an incumbent politician is often explicitly barred from running and must find a new constituency to contest from--a significant disadvantage for an incumbent. We examine whether such adverse redistricting outcomes are less likely to occur for influential incumbents, defined as politicians who are advisory members of the Delimitation Commission, state ministers or members of the ruling party.

We find that India's 2008 redistricting was largely politically neutral. Across the three measures of redistricting and the three indicators of incumbent influence, we find no negative and statistically significant coefficients. In fact, the only significant coefficient is positive, indicating more demographic change in state ministers' constituencies. This is true despite the ministerial asset accumulation "premium" documented in other studies (Fisman et al, 2013). However, with respect to reservation status change, we do see incumbent advisory committee members in eight states and state ministers in four states exerting significant influence, suggesting that some incumbents can influence the process. All findings are robust to alternative model specifications.

We contribute to the redistricting literature by examining the pattern of redistricting and its

¹ The Indian constitution provides for specific, geographically-based "reserved" constituencies in parliament and state legislatures for "Scheduled Castes" (SC) and "Scheduled Tribes" (ST), social groups that were historically marginalized and discriminated against. While many different social groups exist in these constituencies, only members of SC or ST can be candidates in reserved seats.

ex-ante relationship with demographic and political characteristics; the existing literature largely conducts *ex-post* analysis.² We also contribute by conducting *ex-ante* analysis with newly constructed measures that make use of highly detailed GIS maps.³ A third contribution is examining electoral redistricting in a relatively young democracy in the Global South, where voter information and accountability mechanisms are likely less widespread than elsewhere. We also provide policy-relevant findings in advance of India's upcoming re-delimitation.⁴

2. Electoral Redistricting in India

India conducts the largest electoral exercise in the world, with more than 900 million voters and 100 recognized political parties. After independence from colonial rule, electoral redistricting was initially undertaken after each decennial census. However, this stopped after complaints from several states that the process undermined states' incentives to implement population control policies, since a bigger population would result in more state representatives in the national parliament. In 1976, the 42nd Amendment to the Indian Constitution required that all electoral boundaries as of 1977 be frozen in place "until the relevant figures for the first census taken after the year 2000 have been published" (Articles 81 and 82).

In 2002, India began the process of redrawing electoral constituencies following the 2001 census (see Appendix Figure A.1 for the timeline). Since previous concerns regarding population control incentives were still relevant, the delimitation exercise was strictly within-state, i.e. the

² See, among others, Gelman & King (1990); Altman & McDonald (2015); Fraga (2016); *contra* Thompson (2004); Mann (2005); Cain et al (2005); McDonald (2006).

³ Prior literature mostly constructs seats-votes curves for the entire state or country, rather than examining constituency-level changes; Ansolabehere et al. (2000) is a notable exception.

⁴ India's 2021 census was postponed due to the Covid-19 pandemic; it is now planned for 2027.

goal was to equalize within-state constituency sizes and there was no redistribution of national parliament seats across states. Some deviations from perfect equality of population sizes were allowed due to geographical or practical constraints, e.g. a constituency should not include areas from two sides of a river for logistical convenience in conducting elections.

In each state, re-delimitation was carried out by an independent three-member Delimitation Commission, comprising a former Supreme Court judge, the Chief Election Commissioner of India and the State Election Commissioner of the state. Ten elected representatives from the state, five from the state legislature (*Vidhan Sabha*) and five from the national parliament (*Lok Sabha*), acted as “associate members” to advise each state’s Delimitation Commission, though they had no voting power on the Commission’s final decisions. Final reports were published after the publication of draft reports followed by a public comments period (see Appendix A for further details). The new electoral boundaries were approved by the President of India in February 2008. By law, these new boundaries cannot be changed until the first census after the year 2026.

3. Measuring Redistricting

We analyze all 23 states in India that underwent delimitation in 2008,⁵ using village- or town-level demographic data from the 2001 census and overlaying these with GIS maps of pre-2008 (“old”) and post-2008 (“new”) constituency boundaries.⁶ This enables us to compute old and

⁵ For a variety of reasons, the states of Arunachal Pradesh, Assam, Jammu & Kashmir, Jharkhand, Manipur, and Nagaland did not redistrict following the 2001 Census.

⁶ The imperfect GIS overlay was complemented by extensive manual checks. We are unable to match constituencies within cities as the number and boundaries of wards changed substantially. Thus, some large cities (accounting for 6% of our constituencies) are dropped from the analysis.

new constituency demographic characteristics. Our data show that constituency size dispersion was considerably reduced after redistricting (see Appendix Figure A.2). We also see that the extent of change differed considerably across constituencies (see Appendix Figure A.3 for sample maps).

To define the extent of redistricting, we need to match old constituencies to new constituencies; we do this based on whichever new constituency contains the largest proportion of the population from the old constituency. We then construct three measures of the extent of redistricting. Our first measure relates to the extent of demographic change in a given constituency;

it is computed as $\frac{1}{5} \sqrt{\sum_j (D_{j,new} - D_{j,old})^2}$, where $D_{j,old}$ is demographic characteristic j for the pre-

delimitation constituency, and $D_{j,new}$ is the characteristic for the post-delimitation constituency.

The demographic characteristics are the population shares of Scheduled Castes, Scheduled Tribes, women, literate and rural population. Since all the demographic variables are between 0 and 1, this measure would be zero for constituencies where there were no boundary changes and one for constituencies where all the demographics changed from zero to one (e.g. if it went from completely rural to completely urban and from having no SC/ST to having all SC/ST population etc). In our data, the maximum for this measure is 0.210 and the average is 0.017 (Appendix Table B.1). Our second extent of redistricting measure is the fraction of new voters in the new constituency. This is based on the assumption that an incumbent would prefer to retain as many of their old voters as possible. For constituencies with no boundary changes, this measure would be zero and for constituencies where all the voters were new, this measure would be one; in our data, the average is 0.33, i.e. that an incumbent's constituency retains about two-thirds of the original voters. Our third redistricting measure is whether a constituency's reservation status changed. In our data, 13% of unreserved constituencies became reserved post-delimitation, while 7% of reserved constituencies became unreserved.

4. Did Incumbent Politicians Influence Redistricting?

To determine if politicians were able to influence the delimitation process and achieve preferential constituencies, we estimate the following regression:

$$Redistrict_j = \beta_0 + \beta_1 Influence_{1,j} + \beta_2 Influence_{2,j} + \beta_3 Influence_{3,j} + \mathbf{X}_j' \boldsymbol{\gamma} + \eta_j \quad (1)$$

where $Redistrict_j$ is a measure of redistricting in constituency j : demographic change, fraction of new voters, or change in reservation status. $Influence_{1,j}$, $Influence_{2,j}$ and $Influence_{3,j}$ measure the influence of the incumbent state legislator from constituency j . These are, respectively, indicators for whether the incumbent was an advisory member of the Delimitation Commission, a minister in the state government, and a member of the ruling party; all of these were obtained from public sources. Note that these measures are not very highly correlated with each other, beyond the fact that ruling party members are more likely to be state ministers (Appendix Table B.2). We chose each because these are the individuals who would have both an opportunity and an incentive to try to shift the process away from political neutrality. If the redistricting process was particularly favorable to incumbents in the sense of avoiding *unfavorable* redistricting outcomes, we expect the coefficients β_1 through β_3 to be negative and significant. Since the redelimitation process was within-state, we conduct a comparison of constituencies in a state with others within the same state by including a vector of state fixed effects $\mathbf{X}_j$.

We find no evidence that the redistricting process was favorable to influential incumbents. Across our three measures of redistricting, and our three measures of incumbent influence, we find no negative and statistically significant coefficients (Table 1). In fact, the only significant coefficient is positive, suggesting that state ministers' constituencies had a larger extent of demographic change than those of non-ministers.

Table 1: Do incumbent politicians influence redistricting?

	(1) Demographic change	(2) Fraction new voters	(3) Change in reservation status
Incumbent is on delimitation advisory committee	0.002 (0.003)	0.008 (0.024)	-0.009 (0.038)
Incumbent is a state minister	0.004*** (0.002)	0.010 (0.014)	-0.027 (0.021)
Incumbent is a member of the state ruling party	-0.000 (0.001)	-0.005 (0.008)	0.023 (0.014)
Observations	3,251	3,251	3,251
R-squared	0.073	0.039	0.108

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Robust standard errors in parentheses. "Incumbent" refers to the incumbent member of the state legislative assembly from that constituency. Regressions include state fixed effects. Full specification available in Appendix Table B.3.

We verify that these results remain unchanged when we control for pre-redistricting constituency demographic characteristics and how far its population was from the district average, which would shape the extent of boundary change experienced (Appendix Table B.4). The results are also robust to excluding state fixed effects or including state random effects instead of state fixed effects (Table B.5), to including other incumbent characteristics, such as party identity (Table B.6), and to excluding outlier values of the dependent variable (Table B.7). Examining our results state-by-state, we find no incumbent influence on the constituency's demographic change (Figure B.1) and only one or two negative significant coefficients for the fraction of new voters (Figure B.2). When it comes to change in reservation status, we do see incumbent advisory committee members exerting significant influence in eight states and state ministers showing negative and significant coefficients in four states (Figure B.3).

7. Conclusions

This paper examines the 2008 electoral re-delimitation process in India, which substantially changed both state and national electoral constituency boundaries. We find that, by and large, the process achieved its primary goal of reducing population size dispersion across constituencies and that the redistricting process does not appear to have been substantially influenced by incumbent

politicians. In particular, the extent of redistricting is not statistically different for constituencies of ruling party politicians and those in ministerial positions.

Our study constitutes a methodological advance in that it puts forward simple, comparable measures of the extent of constituency redistricting that can easily be computed using GIS data to match constituencies. As such, this methodology is generalizable to other countries and electoral systems. An important policy conclusion is that it is possible to implement politically neutral redistricting plans in a developing country, provided that a non-political body is in charge of the process, and that the process is transparent and inclusive of all relevant stakeholders. India, as well as other countries throughout the world, would do well to replicate this approach in any future delimitation.

5. References

- Altman, M., and M. McDonald (2015): “Redistricting and Polarization,” *American Gridlock*, J.A. Thurber and A. Yoshinaka (eds), Cambridge: Cambridge Univ. Press, 103-131.
- Ansolabehere, S., J. Snyder and C. Stewart (2000): “Old Voters, New Voters and the Personal Vote,” *American Journal of Political Science*, 44(1), 17-34.
- Baqir, R. (2002): “Districting and Government Overspending,” *Journal of Political Economy*, 110(6), 1318-5134.
- Besley, T. and I. Preston (2007): “Electoral Bias and Policy Choice,” *Quarterly Journal of Economics*, 122(4), 1473-1510.
- Cain, B.E. (1985): “Assessing the Partisan Effects of Redistricting,” *American Political Science Review*, 79(2), 320–333.
- Cain, Bruce E., Karin MacDonald, and Michael McDonald (2005): “From Equality to Fairness,” *Party Lines*, T.E. Mann and B.E. Cain (eds), Washington, DC: Brookings Institution.

- Engstrom, E.J. (2006): “Stacking the States, Stacking the House,” *American Political Science Review*, 100(3), 419-427.
- Fisman, R., F. Schulz and V. Vig (2013): “The Private Returns to Public Office,” *Journal of Political Economy*, 122(4), 806-862.
- Forgette, R and J. Winkle (2006): “Partisan Gerrymandering and the Voting Rights Act,” *Social Science Quarterly*, 87(1), 155-173.
- Fraga, B.L. (2016): “Redistricting and the Causal Impact of Race on Voter Turnout,” *Journal of Politics*, 78(1), 19-34.
- Fraga, B.L., D.J. Moskowitz & B. Schneer (2022): “Partisan Alignment Increases Voter Turnout,” *Political Behavior*, 44, 1883–1910.
- Friedman, J. and R. Holden (2009): “The Rising Incumbent Advantage,” *Journal of Politics*, 71(2), 593-611.
- Gelman, A. and G. King (1990): “Estimating the Electoral Consequences of Legislative Redistricting,” *Journal of the American Statistical Association*, 85(410), 274-282.
- Gelman, A. and G. King (1994): “Enhancing Democracy through Legislative Redistricting,” *American Political Science Review*, 88(3), 541-559.
- Gilligan, T.W., and J.G. Matsusaka (1999): “Structural Constraints on Partisan Bias under the Efficient Gerrymander,” *Public Choice*, 100(1-2), 65-84
- Gul, F. and W. Pesendorfer (2010): “Strategic Redistricting,” *American Economic Review*, 100(4), 1616-1641.
- Hayes, D and S. McKee (2009): “The Participatory Effects of Redistricting,” *American Journal of Political Science*, 53(4), 1006-1023.
- Jensenius, Francesca R. (2013): “Was the Delimitation Commission Unfair to Muslims?”, *Studies*

- in Indian Politics*, 1(2), 213-229.
- Mann, T.E. (2005): “Redistricting Reform: What is Desirable? Possible?”, *Party Lines*, T.E. Mann and B.E. Cain (eds), Washington, DC: Brookings Institution.
- McDonald, M.P. (2006): “Redistricting and Competitive Districts,” *The Marketplace of Democracy*, M.P. McDonald and J. Samples (eds), Washington, DC: Brookings Institute & Cato Institute, 222–224.
- McMillan, A. (2000): “Delimitation, democracy, and end of constitutional freeze,” *Economic and Political Weekly*, 35(15), 1271-1276.
- Mishra, R.K. (2025): *Demography, Representation, Delimitation*, Chennai, India: Westland.
- Patel, P.L, and T.V. Sekher (2024): “Parliamentary Delimitation,” *Journal of Asian and African Studies*, 1-16.
- Rossiter, D.J., R.J. Johnston, and C.J. Pattie (1997): “Estimating the Partisan Impact of Redistricting in Great Britain,” *British Journal of Political Science*, 27(2), 299–331.
- Shotts, K.W. (2001): “The Effect of Majority-Minority Mandates on Partisan Gerrymandering,” *American Journal of Political Science*, 45(1), 120-135.
- Singh, C.P. (2000): “A Century of Constituency Delimitation in India,” *Political Geography*, 19(4), 517-532.
- Thomas, A.C. (2011): “Avoiding an Electoral Lost Decade,” *SSRN Working Paper*.
- Thompson, D.F. (2004): “Election Time,” *American Political Science Review*, 98(1), 51–63.
- Verma, A.K. (2002): “Issues and Problems in India’s Delimitation Exercise,” *The Indian Journal of Political Science*, 63(4), 371–388.
- Williamson, R.D. (2019): “Examining the Effects of Partisan Redistricting on Candidate Entry Decisions,” *Election Law Journal: Rules, Politics & Policy*, 18(3), 214-226.

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Appendix A: Electoral Redistricting in India

India's Political System

India's political system is a parliamentary democracy at both the national and state levels, with first-past-the-post elections conducted every five years. State elections have an average of ten candidates contesting from each constituency. Voter turnout is quite high: it was approximately 58% in the 2009 general election and 66.1% in the most recent general elections, in 2024. In 2001, Scheduled Castes (SCs) and Scheduled Tribes (STs) accounted for 24.4% of the population, an increase from 21.5% in the 1971 census. These groups are provided reserved constituencies in the national parliament and state legislatures, in proportion to their population share. Prior to re-delimitation, 22.1% of national parliamentary seats were reserved for these communities, and 24.1% were reserved after re-delimitation.

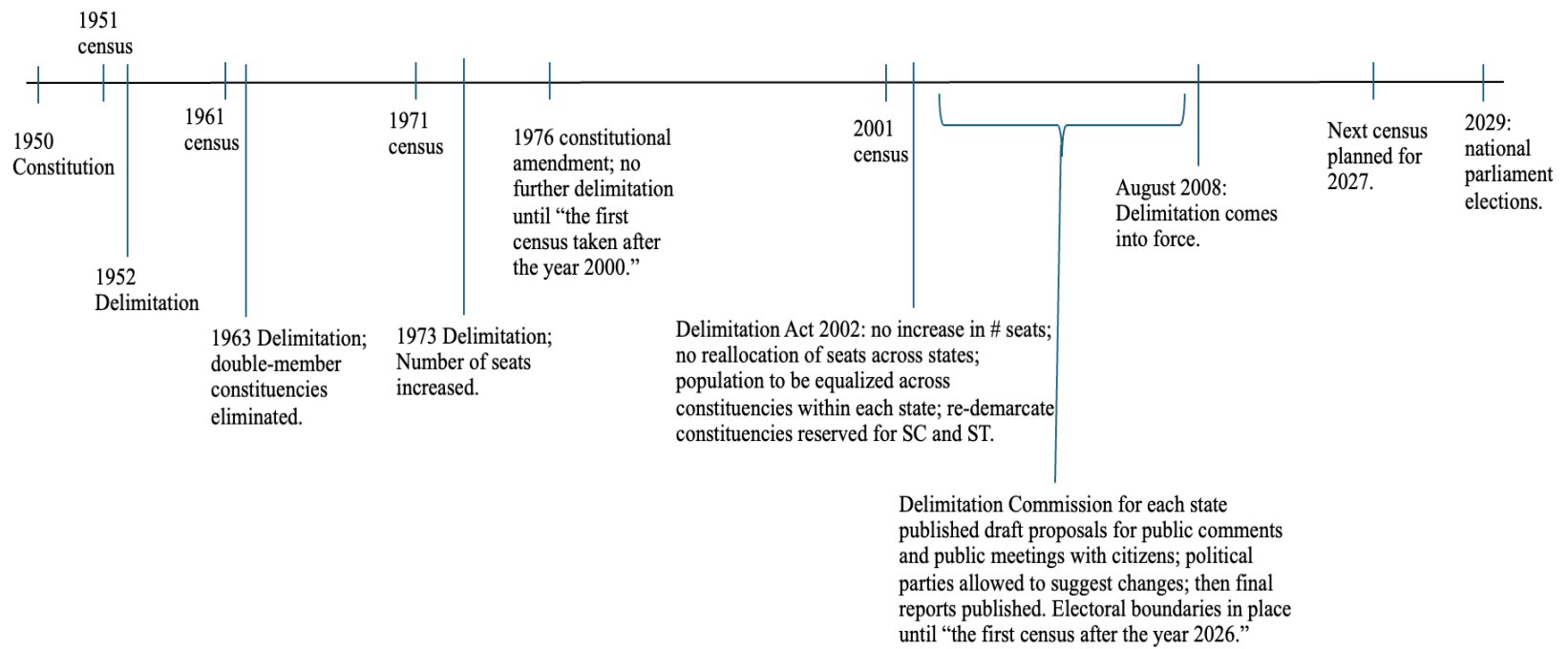
The Election Commission of India is responsible for conducting elections to the national parliament and to state legislative assemblies. Election Commission staff are career bureaucrats who are required to be politically neutral. For instance, they may not join political parties or otherwise engage in partisan political activity. The Election Commission of India has established a reputation for conducting free and fair elections and has undertaken many measures to safeguard the integrity of elections (McMillan, 2010; Ahuja & Ostermann 2018). Voting in India has been completely electronic since 2004, and elections are often held in several phases to ensure adequate security arrangements.

Delimitation after the census of 2001

As described earlier, electoral redistricting in India occurred after every decennial census until concerns about conflicting with population control policies resulted in a freezing of electoral boundaries in 1976 until the first census after the year 2000 (see timeline in Figure A.1). After the census of 2001, the delimitation process commenced began with the enactment of the Delimitation Act, 2002 and the Delimitation (Amendment) Act 2003. Since the concerns about distorting population control incentives for state governments continued to be relevant, this redistricting exercise specified that the total number of electoral constituencies would remain the same, both for the national legislature as well as for state legislatures. Further, each state would continue to have the same number of representatives in the national legislature, i.e. there would be no reallocation of seats across states. The decision not to increase the number of state representatives in line with increases in state population, and the decision to not reallocate parliamentary seats across states, have both come under heavy criticism as violating the principles of representative democracy (McMillan 2001; Yadav 2008).

The goals of this redistricting exercise were two-fold: First, to equalize the population across electoral constituencies *within* each state, and second, to re-demarcate the electoral constituencies to be reserved for the SCs and STs in proportion to their increased population share since the 1971 census. Since this was the first redrawing of electoral boundaries in three decades, it resulted in widespread changes to the electoral boundaries. Rural-urban migration had resulted in much faster growth of urban population compared to rural population: 19.9% of India's population lived in urban areas in 1971, compared to 27.8% in 2001. The redistricting exercise therefore resulted in a greater allocation of electoral seats to urban areas.

Appendix Figure A.1: Delimitation Timeline



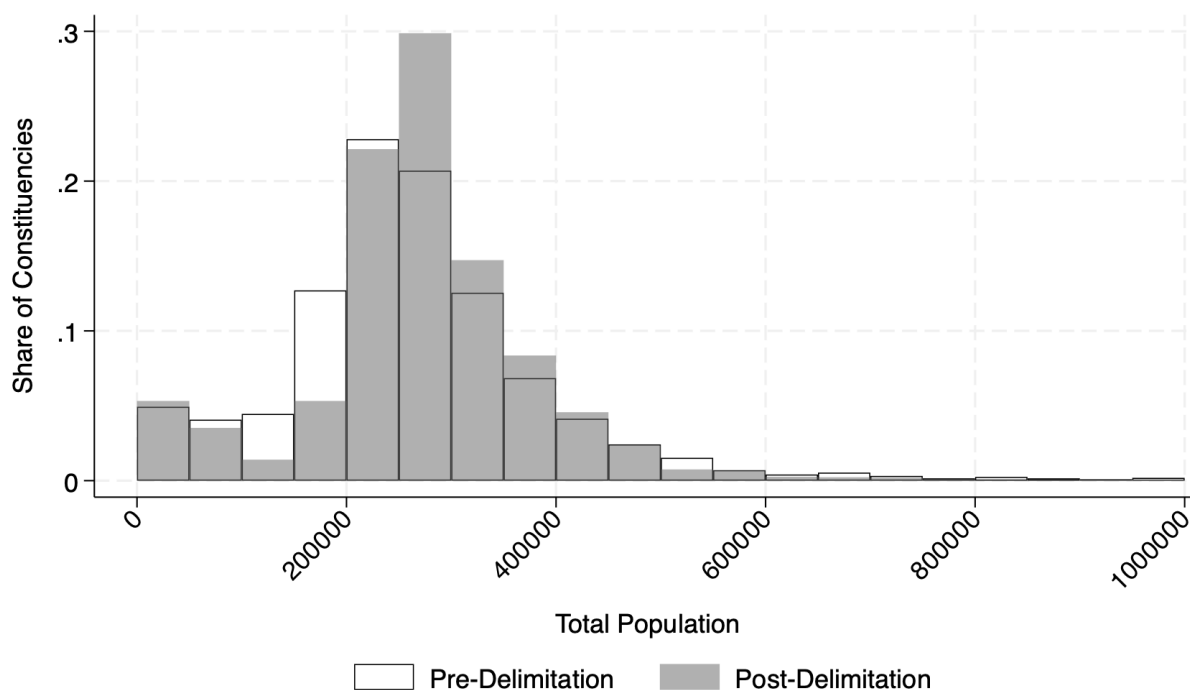
As described earlier, the Delimitation Commission for each state comprised a retired Supreme Court judge, the Chief Election Commissioner of India and the Chief Election Commissioner of the state, together with an advisory body of incumbent politicians. After examining data from the 2001 census and local maps, consulting with district officials, and meeting with these associate members, the Delimitation Commission prepared a detailed draft proposal with the proposed boundaries of each electoral constituency. This draft proposal was widely published, public comments were invited, and public meetings in one or more places were held to hear the views of citizens. Political parties in the state scrutinized these proposals and submitted their views for consideration, often proposing new boundaries for certain constituencies. For instance, the associate members of the advisory committee in Andhra Pradesh proposed changes to more than 15 state assembly constituency boundaries after the draft proposals were published and requested that their dissent be published along with the proposal (Delimitation Commission of India, 2007). After taking into account these views, the Delimitation Commission published final reports for each state.

The explicit goal of the Delimitation Commission was to redraw constituencies such that “the population of each parliamentary and assembly constituency in a State shall, so far as practicable, be the same throughout the State” (Delimitation Commission of India, 2004). This was subject to the constraints that the constituencies should be geographically compact and contiguous, every state assembly constituency should lie wholly within a national parliamentary constituency, and all assembly constituencies should lie wholly within administrative districts. Factors such as physical features, facilities of communication and public convenience were also to be considered, such that areas divided by rivers or hilly ranges or forests or ravines and other such natural barriers were not to be put in the same constituency. This consideration is mainly

related to the logistics of conducting elections within the constituency. Since these constraints make the exact equalization of population across constituencies difficult, the Delimitation Commission agreed that the population in a specific constituency could vary up to 10 percent above or below the district average. This guideline has been criticized for being arbitrary, and for being violated in many cases (Kumar 2009).

Figure A.2 shows the distribution of constituency populations before and after delimitation. We see that the post-delimitation population distribution is more concentrated towards the center of the distribution i.e. there is less variation in constituency sizes after the delimitation. The variance of constituency populations prior to delimitation was 163,035 which decreased to 127,845 after delimitation (a 22% decline). Thus, the delimitation process achieved the primary goal of considerable (though not complete) equalization of population sizes across constituencies.

Figure A.2: Redelimitation resulted in more equal population sizes across constituencies



Note: Constituencies with populations over 1,000,000 people were excluded from this figure. This includes 6 constituencies in the pre-delimitation sample and 17 from the post-delimitation sample.

After these constituencies were mapped out, constituencies that were to be reserved for the Scheduled Castes (SCs) and Scheduled Tribes (STs) were demarcated. The overall number of constituencies to be reserved for the SCs and STs were based on their population share in the state, and the exact constituencies chosen for reservation were the ones that have the largest population shares of these communities. In our analysis sample of 3251 state assembly constituencies, there were 492 constituencies reserved for SC and 237 reserved for ST prior to redelimitation; these numbers increased to 563 and 366 respectively after redelimitation, consistent with the increase in SC and ST population share.

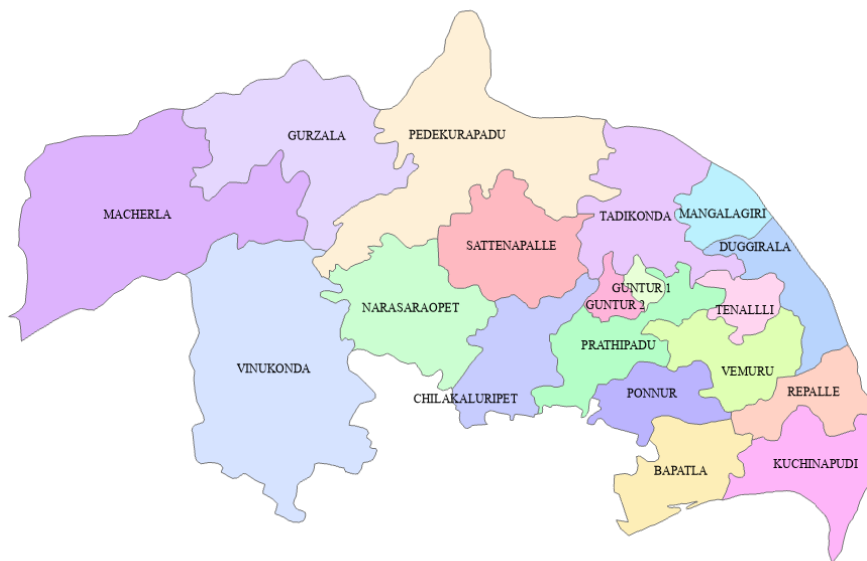
Electoral redistricting has generated significant politician reactions and media attention, but little empirical analysis. There was media speculation that the two largest national parties, the Indian National Congress (INC) and the Bharatiya Janata Party (BJP), might even join hands to delay the national implementation of the final report, although this did not come to pass (Das 2006). Some media reports claimed that many politicians' electoral prospects were harmed by this process (*The Hindu* 2009; Rahman 2013), but other commentators cited the Delimitation Commission's work as being "without any obvious political partisanship" (*Economic and Political Weekly* 2008).

To our knowledge, there has been no systematic empirical analysis of the redistricting process. In particular, it should be noted that different constituencies underwent different degrees of change. Appendix Figure A.3 shows the old and new constituency maps of Guntur district in Andhra Pradesh. It is noticeable that the boundaries of some constituencies did not change at all (e.g. Macherla and Vinukonda), some became larger (e.g. Sattenapalle) while yet others became smaller than before (e.g. Prathipadu). The shapes of many constituencies have changed, as have

their names. Also note that Guntur had 19 constituencies before redistricting but only 17 after redistricting.

Figure A.3: Pre- and post-redistricting constituency boundaries in Guntur district

Panel A: Pre-redistricting constituencies



Panel B: Post-redistricting constituency boundaries



Appendix References

- Ahuja, A. and S.L. Ostermann (2018): "From Quiescent Bureaucracy to Undocumented Wonder," *Governance*, 31(4), 759-776.
- Das, A.K. (2006): "Delhi's Political Map Prepares to Get a Makeover," *The Times of India*, September 17, 2006, p. 2.
- Delimitation Commission of India (2004): "Delimitation of Assembly and Parliamentary Constituencies-Guidelines and Methodology," No. 282/DEL/2004, March 17, 2004.
- Delimitation Commission of India (2007): "Order No. 37," *Gazette of India*, Part II, Section 3 (iii), January 22, 2007.
- Delimitation Commission of India (2008): "Changing Face of Electoral India: Delimitation 2008," Delimitation Commission of India, New Delhi: Gov. Press.
- Election Commission of India (1976). "Delimitation of Parliamentary and Assembly Constituencies Order, 1976." Election Commission of India, New Delhi: Gov. Press.
- Economic and Political Weekly* (Editorial, June 14, 2008): "Delimitation: A Tale of Missed Opportunities," *Economic and Political Weekly*, 43(24).
- Kumar, Sanjay (2009): "The Fourth Delimitation: An Evaluation," *Economic and Political Weekly*, 44(3), 41-46.
- McMillan, A. (2001): "Population Change and the Democratic Structure," *Seminar* 506.
- McMillan, A. (2010): "The Election Commission," *The Oxford Companion to Politics in India*, N.G. Jayal and P.B. Mehta (eds), Oxford: Oxford University Press.
- Rahman, N.A. (2013): "Wary BJP Strives to Reverse Effects of Delimitation," *The Times of India*, October 4, 2013.

The Hindu (March 22, 2009): “Delimitation Hits 11 Congress MLAs,”

<http://www.hindu.com/2009/03/22/stories/2009032260670600.htm>, last accessed

November 2013.

Yadav, Y. (2008): “The Paradox of Political Representation,” *Seminar* 586.

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Appendix B: Additional Tables and Figures

Appendix Table B.1: Summary statistics

	(1) Observations	(2) Mean	(3) Standard deviation	(4) Minimum	(5) Maximum
<u>Panel A: Demographic Characteristics</u>					
Population	3,251	264,209	163035	314	3,190,943
Fraction literate	3,251	0.540	0.132	0	0.868
Fraction women	3,251	0.487	0.016	0.404	0.566
Fraction SC	3,251	0.170	0.094	0	0.946
Fraction ST	3,251	0.109	0.212	0	1
Fraction rural	3,251	0.824	0.224	0	1
<u>Panel B: Main Outcome Variables</u>					
Demographic Change	3,251	0.017	0.021	0	0.210
Fraction of New Voters	3,251	0.329	0.224	0	0.999
Reservation Change	3,251	0.199	0.399	0	1
<u>Panel C: Measures of influence</u>					
Incumbent is on delimitation advisory committee	3,251	0.030	0.169	0	1
Incumbent is a state minister	3,251	0.112	0.315	0	1
Incumbent is a member of the state ruling party	3,251	0.478	0.500	0	1

Panel A shows demographic characteristics of village-level data from the 2001 population census, aggregated to pre-delimitation assembly constituencies. Panel B shows measures of redistricting. Measures of influence are for incumbent members of the state legislative assembly.

Appendix Table B.2: Correlation between measures of political influence

	Incumbent is on delimitation advisory committee	Incumbent is a state minister	Incumbent is a member of the state ruling party
Incumbent is on delimitation advisory committee	1		
Incumbent is a state minister	0.036**	1	
Incumbent is a member of the state ruling party	0.015	0.181***	1

Note: *** p<0.01, ** p<0.05, * p<0.1.

Appendix Table B.3: Do Incumbent Politicians Influence Redistricting? (All Coefficients)

	(1) Demographic change	(2) Fraction new voters	(3) Change in reservation status
Incumbent is on delimitation advisory committee	0.002 (0.003)	0.008 (0.024)	-0.009 (0.038)
Incumbent is a state minister	0.004*** (0.002)	0.010 (0.014)	-0.027 (0.021)
Incumbent is a member of the state ruling party	-0.000 (0.001)	-0.005 (0.008)	0.023 (0.014)
Bihar Fixed Effect	-0.007*** (0.001)	0.053*** (0.018)	-0.027 (0.031)
Chhattisgarh Fixed Effect	0.006* (0.003)	0.100*** (0.027)	-0.005 (0.045)
Goa Fixed Effect	0.019*** (0.005)	0.143*** (0.043)	-0.071 (0.053)
Gujarat Fixed Effect	0.004** (0.002)	0.089*** (0.022)	-0.034 (0.035)
Haryana Fixed Effect	0.008*** (0.003)	0.123*** (0.026)	0.073 (0.051)
Himachal Pradesh Fixed Effect	-0.006*** (0.002)	0.005 (0.027)	-0.027 (0.047)
Karnataka Fixed Effect	0.002 (0.002)	0.046** (0.021)	0.078** (0.038)
Kerala Fixed Effect	0.009*** (0.003)	0.133*** (0.025)	-0.062* (0.035)
Madhya Pradesh Fixed Effect	0.003* (0.002)	0.001 (0.020)	0.018 (0.035)
Maharashtra Fixed Effect	0.003 (0.002)	0.065*** (0.020)	-0.002 (0.033)
Meghalaya Fixed Effect	0.012*** (0.004)	0.140*** (0.037)	0.772*** (0.044)
Mizoram Fixed Effect	0.012** (0.006)	-0.004 (0.046)	0.808*** (0.041)
Orissa Fixed Effect	-0.002 (0.002)	0.047** (0.022)	0.070* (0.042)
Puducherry Fixed Effect	0.005 (0.008)	0.083 (0.071)	0.001 (0.086)
Punjab Fixed Effect	0.002 (0.002)	0.107*** (0.026)	0.124** (0.050)
Rajasthan Fixed Effect	-0.004** (0.001)	0.046** (0.020)	0.011 (0.036)
Sikkim Fixed Effect	0.002 (0.003)	0.116** (0.045)	0.030 (0.075)

Tamil Nadu Fixed Effect	0.002 (0.002)	0.078*** (0.019)	-0.000 (0.034)
Tripura Fixed Effect	0.017*** (0.003)	0.201*** (0.039)	0.048 (0.058)
Uttar Pradesh Fixed Effect	-0.004*** (0.001)	0.094*** (0.018)	0.069** (0.031)
Uttarakhand Fixed Effect	-0.008*** (0.001)	0.065** (0.031)	0.041 (0.053)
West Bengal Fixed Effect	0.001 (0.002)	0.037* (0.019)	-0.008 (0.032)
Observations	3,251	3,251	3,251
R-squared	0.073	0.039	0.108

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Robust standard errors in parentheses. "Demographic change" is an index of the difference in demographic characteristics of the old and the new constituency.

"Fraction new voters" is the share of voters in the new constituency who were not in the incumbent's old constituency. "Change in reservation status" equals one if an unreserved constituency (pre-2008) becomes reserved for Scheduled Castes or Scheduled Tribes (post-2008), or vice versa. "Incumbent" refers to the incumbent member of the state legislative assembly.

Appendix Table B.4: Robustness to inclusion of additional controls

	(1) Demographic change	(2) Fraction new voters	(3) Change in reservation status
Incumbent is on delimitation advisory committee	0.002 (0.003)	0.009 (0.024)	-0.005 (0.036)
Incumbent is a state minister	0.004*** (0.001)	0.014 (0.013)	-0.022 (0.021)
Incumbent is a member of the state ruling party	-0.000 (0.001)	-0.005 (0.008)	0.014 (0.014)
Absolute Value of Population Deviation from District Average	0.002*** (0.001)	-0.009** (0.004)	0.007 (0.006)
Fraction of Population that is Literate - Old AC	0.008 (0.006)	0.204*** (0.057)	0.037 (0.083)
Fraction of Population that is SC - Old AC	0.018* (0.010)	0.041 (0.069)	1.186*** (0.114)
Fraction of Population that is ST - Old AC	0.009*** (0.003)	-0.090*** (0.032)	0.156*** (0.048)
Fraction of Population that is Rural - Old AC	-0.012*** (0.003)	0.250*** (0.026)	0.028 (0.035)
Fraction of Women in Population - Old AC	-0.076*** (0.029)	1.154*** (0.341)	-0.377 (0.541)
Observations	3,251	3,251	3,251
R-squared	0.119	0.109	0.150

Note: *** p<0.01, ** p<0.05, * p<0.1. Robust standard errors in parentheses. "Demographic change" is an index of the difference in demographic characteristics of the old and the new constituency. "Fraction new voters" is the share of voters in the new constituency who were not in the incumbent's old constituency. "Change in reservation status" equals one if an unreserved constituency (pre-2008) becomes reserved for Scheduled Castes or Scheduled Tribes (post-2008), or vice versa. "Incumbent" refers to the incumbent member of the state legislative assembly.

Appendix Table B.5: Robustness to excluding state fixed effects or including state random effects

	No State Fixed Effects			With State Random Effects		
	(1) Demographic change	(2) Fraction new voters	(3) Change in reservation status	(4) Demographic change	(5) Fraction new voters	(6) Change in reservation status
Incumbent is on delimitation advisory committee	0.005 (0.003)	0.023 (0.025)	0.039 (0.044)	0.003 (0.003)	0.011 (0.023)	-0.005 (0.036)
Incumbent is a state minister	0.006*** (0.002)	0.025* (0.014)	0.043* (0.024)	0.005*** (0.002)	0.014 (0.018)	-0.023 (0.026)
Incumbent is a member of the state ruling party	0.000 (0.001)	-0.012 (0.008)	0.005 (0.014)	-0.000 (0.001)	-0.005 (0.007)	0.021 (0.017)
Observations	3,251	3,251	3,251	3,251	3,251	3,251
R-squared	0.010	0.002	0.002	0.009	0.002	0.000

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Robust standard errors in parentheses. "Demographic change" is an index of the difference in demographic characteristics of the old and the new constituency. "Fraction new voters" is the share of voters in the new constituency who were not in the incumbent's old constituency. "Change in reservation status" equals one if an unreserved constituency (pre-2008) becomes reserved for Scheduled Castes or Scheduled Tribes (post-2008), or vice versa. "Incumbent" refers to the incumbent member of the state legislative assembly.

Appendix Table B.6: Robustness Check - Additional Measures of Influence

	(1)	(2)	(3)
	Demographic Change	Fraction New Voters in New Constituency	Change in Reservation Status of Constituency
Incumbent is on delimitation advisory committee	0.002 (0.003)	0.008 (0.024)	-0.008 (0.038)
Incumbent is a state minister	0.004*** (0.002)	0.009 (0.014)	-0.028 (0.021)
Incumbent is a member of the state ruling party	-0.000 (0.001)	-0.006 (0.008)	0.022 (0.014)
MP is on Delimitation Committee	-0.001 (0.001)	-0.006 (0.011)	-0.009 (0.020)
Incumbent is a member of the BJP	0.003** (0.001)	0.015 (0.012)	0.003 (0.022)
Incumbent is a member of the INC	0.001 (0.001)	-0.048*** (0.018)	-0.040 (0.030)
Incumbent belongs to party of the 2008 national ruling coalition	-0.001 (0.001)	0.035** (0.017)	0.027 (0.028)
Observations	3,251	3,251	3,251
R-squared	0.074	0.042	0.109

Note: *** p<0.01, ** p<0.05, * p<0.1. Robust standard errors in parentheses. "Demographic change" is an index of the difference in demographic characteristics of the old and the new constituency. "Fraction new voters" is the share of voters in the new constituency who were not in the incumbent's old constituency. "Change in reservation status" equals one if an unreserved constituency (pre-2008) becomes reserved for Scheduled Castes or Scheduled Tribes (post-2008), or vice versa. "Incumbent" refers to the incumbent member of the state legislative assembly.

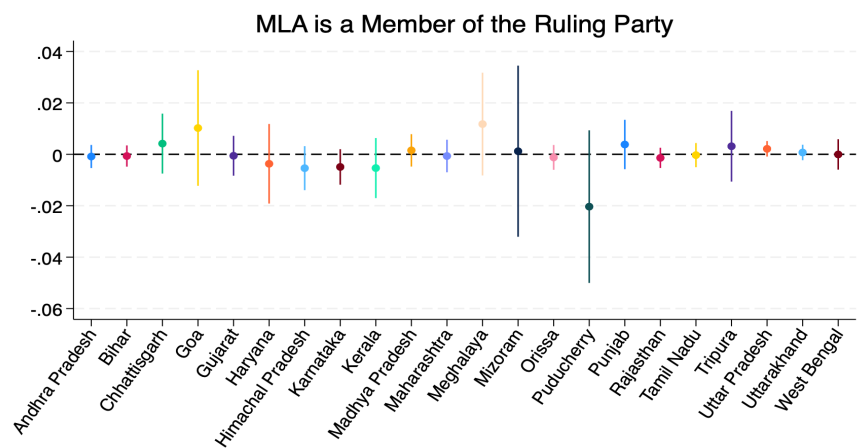
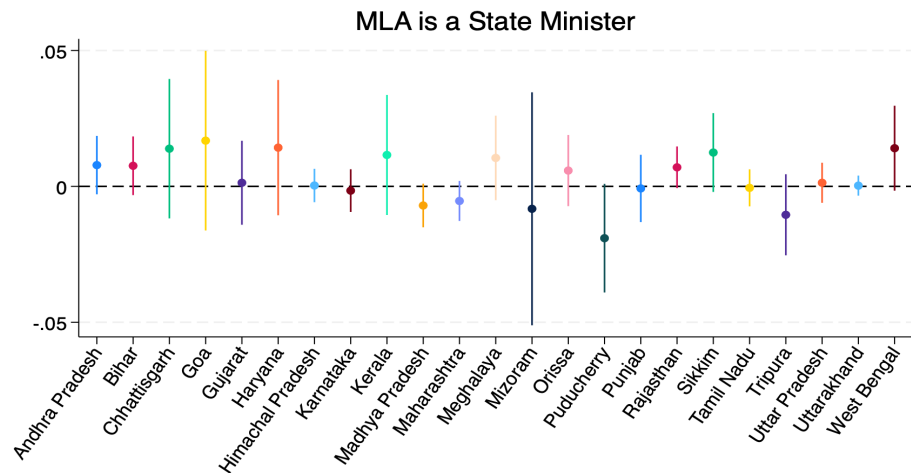
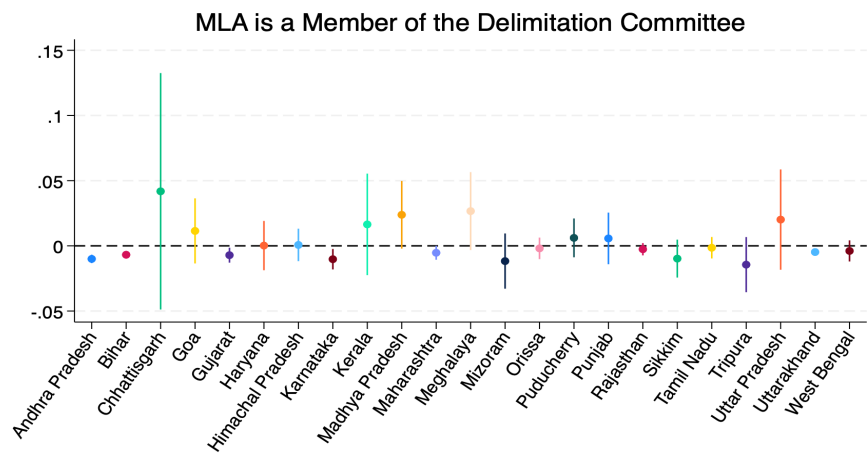
Appendix Table B.7: Robustness Check - Dropping Extreme Values

	(1) Demographic change	(2) Fraction new voters
Incumbent is on delimitation advisory committee	-0.001 (0.001)	-0.009 (0.020)
Incumbent is a state minister	-0.001 (0.001)	-0.002 (0.012)
Incumbent is a member of the state ruling party	-0.000 (0.000)	-0.010 (0.007)
Observations	2,925	2,925
R-squared	0.069	0.027

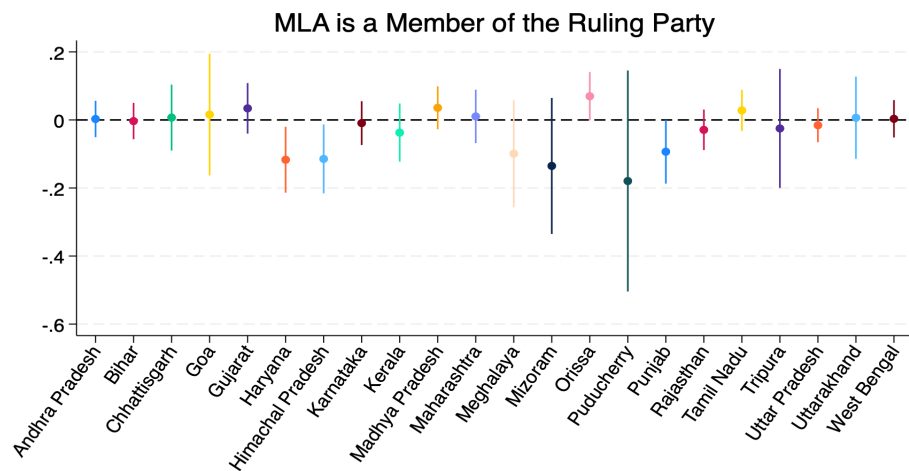
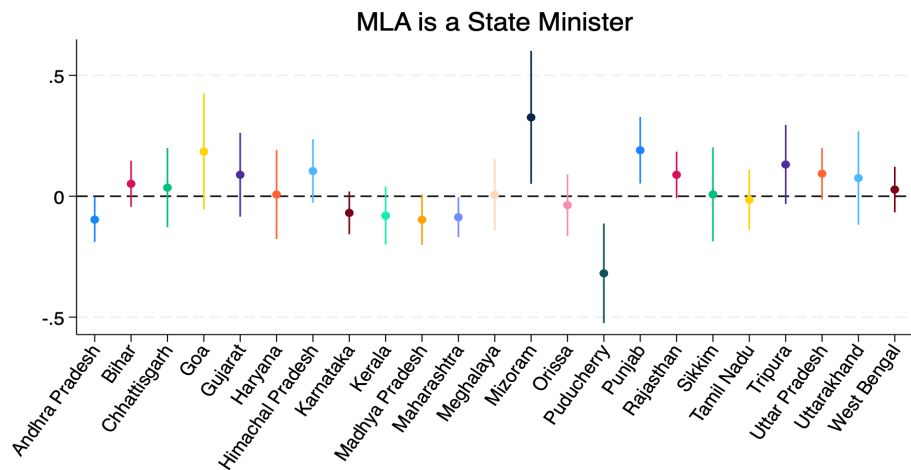
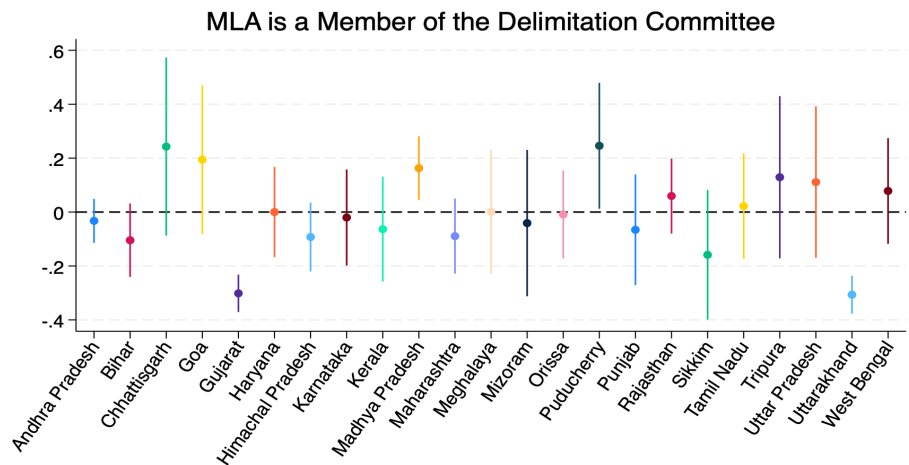
Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Robust standard errors in parentheses. "Demographic change" is an index of the difference in demographic characteristics of the old and the new constituency.

"Fraction new voters" is the share of voters in the new constituency who were not in the incumbent's old constituency. "Change in reservation status" equals one if an unreserved constituency (pre-2008) becomes reserved for Scheduled Castes or Scheduled Tribes (post-2008), or vice versa. "Incumbent" refers to the incumbent member of the state legislative assembly. Column 1 drops constituencies in the top 10 percent of values of the demographic change measure. Column 2 drops constituencies in the top 10 percent of the fraction of new voters variable.

Appendix Figure B.1: Demographic Change Measure - State-By-State Variation



Appendix Figure B.2: Fraction of New Voters in New Constituency - State-By-State Variation



Appendix Figure B.3: Reservation Status Change - State-By-State Variation

